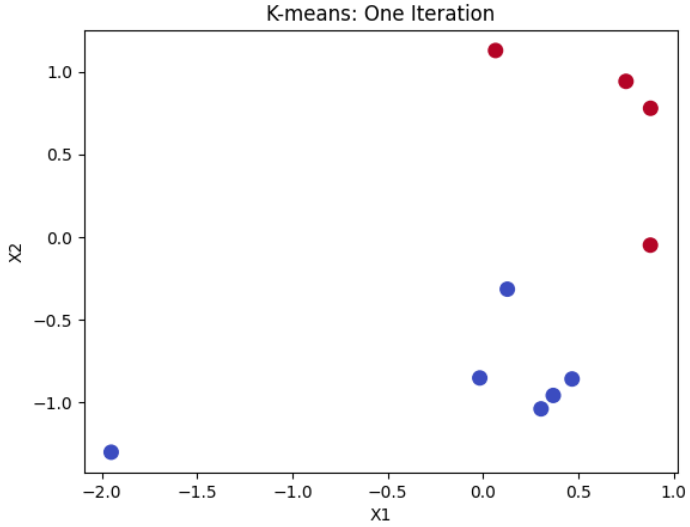
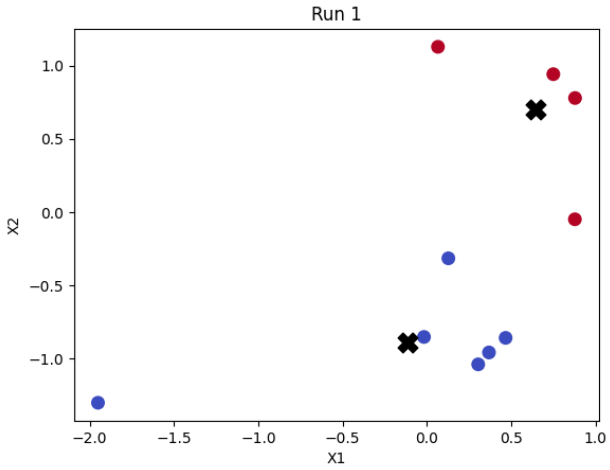
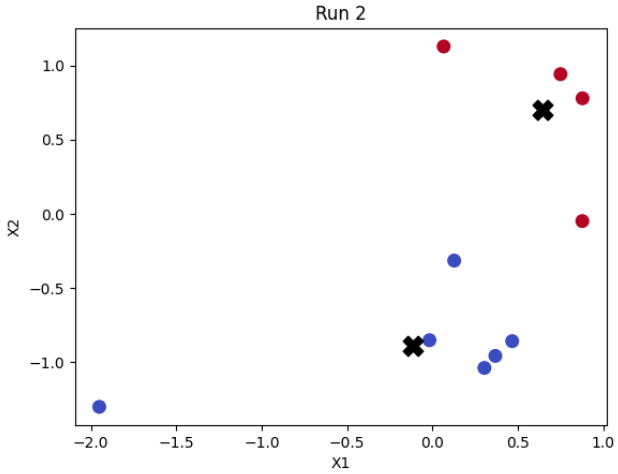
Question 1: Implementing K-means

Report the means and updated assignments, and plot the observations colored by their updated assignments.
Seed: 42
Means: [[-0.0829941 -0.67525964]
 [0.45805639 0.16846899]]
Initial: [1 2 1 2 1 1 2 2 1 2]
Updated: [1 2 1 1 1 2 2 1 1 2]



Recalculate the cluster means from the final assignments before returning them. Return the final assignments, cluster means, and within-cluster sum of squares





Run 1 means: [[-0.11650312 -0.88827087]
 [0.64358254 0.69891799]]
Run 1 WCSS: 5.98234186277417
Run 2 means: [[-0.11650312 -0.88827087]
 [0.64358254 0.69891799]]
Run 2 WCSS: 5.98234186277417

Report the final plots, cluster means, and within-cluster sums of squares. Do the two runs produce the same grouping and objective value? Explain your finding

The two runs started with different random assignments but converged to the same final clustering, with identical cluster means and WCSS values of 5.9823. This shows that different initializations do not necessarily produce different results. However, K-means can converge to different local optima depending on the initial assignments.

Question 2: Image color compression

Plot the original image and report its dimensions. Transform it into a data matrix with one observation per pixel and three variables: red, green, and blue. Report the dimensions of this matrix.

Image dimensions: (112, 200, 3)
Data matrix dimensions: (22400, 3)



For $K = 2,3,5,10$, apply your function to the pixel data. Report the cluster means and within-cluster sum of squares for each K. Reconstruct and plot each compressed image by replacing every pixel with its assigned cluster mean. Use the same RGB scale for all fits.

K=2, WCSS=1547.4807
RGB cluster means:
 [[0.764 0.875 0.9]
 [0.344 0.562 0.634]]

K=3, WCSS=669.1549

RGB cluster means:
[[0.965 0.926 0.9]
[0.371 0.456 0.463]
[0.45 0.763 0.868]]

K=5, WCSS=275.8361
RGB cluster means:
[[0.525 0.821 0.909]
[0.302 0.662 0.801]
[0.987 0.952 0.936]
[0.324 0.413 0.428]
[0.754 0.692 0.61]]

K=10, WCSS=153.7076
RGB cluster means:
[[0.52 0.819 0.908]
[0.359 0.633 0.72]
[0.26 0.676 0.853]
[0.69 0.881 0.941]
[0.238 0.386 0.438]
[0.814 0.743 0.655]
[0.936 0.861 0.847]
[0.995 0.964 0.948]
[0.435 0.419 0.372]
[0.621 0.58 0.51]]

Which value of *K*would you choose for this image? Briefly justify your choice using the reconstructed images and the within-cluster sums of squares.
I would choose K = 3 because the reconstructed image preserves the characters' main features and enough visual detail. Its WCSS is 669.15, which is much lower than K = 2 (1547.48). Although larger K values further reduce WCSS, K = 3 provides a reasonable balance between image quality and the number of colors.



Question 3: Hierarchical clustering of cars

Plot the three dendrograms with the car names and compare the results.
Choose one linkage method and a number of clusters that you consider reasonable.
Single linkage shows a chaining effect, where cars are gradually merged into larger clusters, making the groups less clearly separated.

Complete linkage produces more distinct and compact clusters, with larger distances between major groups.
Average linkage provides a balance between the two methods, showing clearer cluster separation than single linkage but less extreme separation than complete linkage.

I would choose **complete linkage** because its dendrogram shows relatively clear separation between groups, making it easier to select a reasonable number of clusters, such as K = 3.

Based on the clustering, the three groups appear to represent different types of cars: lighter, fuel-efficient cars; heavier, more powerful cars; and cars with moderate performance. The cluster means table can confirm these differences by comparing MPG, horsepower, and weight.

Standardization is important because the variables have different scales. Without standardization, variables with larger numerical values, such as horsepower, could dominate the Euclidean distances and influence the clustering results.

